

Indian Statistical Institute
M.Tech. (CS), First Year, Final of Second Semester Examination, 2021-22
Database Management Systems

Full Marks: 50

Date: 08-06-2022

Time: 2.5 Hours

Answer any *five* of the following questions

$5 \times 10 = 50$

1. Prove the following equivalences between relational algebra expressions:

(a) $(R_1 \bowtie_{\theta_1} R_2) \bowtie_{\theta_2 \wedge \theta_3} R_3 \equiv R_1 \bowtie_{\theta_1 \wedge \theta_3} (R_2 \bowtie_{\theta_2} R_3)$. Note that θ_2 involves attributes only from R_2 and R_3 .

(b) $\sigma_{\theta_1 \wedge \theta_2}(R) \equiv \sigma_{\theta_1}(\sigma_{\theta_2}(R)) \equiv \sigma_{\theta_2}(\sigma_{\theta_1}(R))$.

5+5

2. (a) Given a pair of relations $R_1 = \langle X, Y \rangle$ and $R_2 = \langle Y, Z \rangle$ with 3000 and 2000 tuples, respectively. Let the ranges of values for X and Y in R_1 be $[0, 40)$ and $[0, 25)$, respectively and for Y and Z in R_2 are $[5, 25)$ and $[0, 30)$, respectively. Estimate the size of the following query.

$$\sigma_{X < 10 \wedge Z > 10}(R_1 \bowtie R_2)$$

(b) Show two different cases wherein hash join is better than sort merge join and vice versa.

6+(2+2)

3. In India, Panchayati Raj System functions hierarchically in three levels. We have, in the lowest level of the hierarchy, Gram Panchayat, then Panchayat Samiti, and finally Zila Parishad. They work at the village level, block level and district level, respectively. Consider that a District Magistrate maintains a database of the names of different people in different posts of different types — selected (Government recruited) or elected (chosen through election). The following schema represents the association between the different tables.

VILLAGE = $\langle \underline{vill-name}$: string, $post$: string, $type$: string, $name$: string, $bloc-name$: string $\rangle$

BLOCK = $\langle \underline{bloc-name}$: string, $post$: string, $type$: string, $name$: string, $dist-name$: string $\rangle$

DISTRICT = $\langle \underline{dist-name}$: string, $post$: string, $type$: string, $name$: string $\rangle$

The primary keys are underlined in the corresponding tables.

(a) What will be returned by the following query?

select DISTRICT. $name$ from VILLAGE, BLOCK, DISTRICT where DISTRICT. $dist-name$ = BLOCK. $dist-name$ and BLOCK. $bloc-name$ = VILLAGE. $bloc-name$ and VILLAGE. $name$ like '% Tripathi' and VILLAGE. $type$ = 'selected';

(b) Suppose the logical plan for the query given above is prepared based on the following relational algebra expression.

$\pi_{\text{DISTRICT.name}}((\sigma_{\text{VILLAGE.name like '% Tripathi' and VILLAGE.type = 'selected'}}(\text{VILLAGE}) \bowtie$
 $(\text{BLOCK} \bowtie \text{DISTRICT}))$

Propose a logical query plan that is better than this. Under what circumstances the plan suggested by you is better? What will be the best physical plan for the join operations?

3+(3+2+2)

4. Consider a relational schema $R = \langle M, B, C, D, E, F, G, H, I, J, K, Z, A \rangle$, where all the fields in R contain atomic values. Let $F = \{M \rightarrow BCDE, E \rightarrow FGH, I \rightarrow J, MI \rightarrow K, MZ \rightarrow A\}$ be a set of functional dependencies such that F^+ is exactly the set of functional dependencies that hold for R .
- Find out the candidate keys of R .
 - Perform BCNF decomposition on R until all the redundancies are removed.
 - Find out the largest normal forms of all the decomposed relations obtained in (b).

2+5+3

5. Let there be a relation with data items A, B and C , and 4 different transactions working on them simultaneously at a particular period of time. Suppose a timestamp-based protocol is used to manage their concurrency. How many minimum transactions must be rolled back for satisfying the consistency if a system crash happens after the timestamp 13. What transactions are to be precisely rolled back and why?

Transaction T_1	Transaction T_2	Transaction T_3	Transaction T_4	Timestamp
read(A)				1
write(A)				2
	read(C)			3
	write(C)			4
			write(A)	5
	read(A)			6
		read(B)		7
		write(B)		8
	write(A)			9
	read(B)			10
write(B)				11
commit				12
	write(B)			13
		commit		14
	commit			15
			read(C)	16
			write(C)	17
			commit	18

2+(4+4)

6. Give appropriate justifications supporting the following statements:
- Range searching is more efficient on B^+ -trees than using a hash table.
 - BCNF decomposition cannot preserve functional dependency.
 - Topological sorting on an acyclic precedence graph will yield the serializability order.

- (d) Tree-based concurrency control protocols cannot allow shared locks on data items.
- (e) Recovery in immediate database modification is faster than deferred database modification.

2+2+2+2+2

7. Give appropriate examples of the following things with justifications:

- (a) An E-R diagram having a relationship set defined on an entity set with itself.
- (b) A relation having multiple superkeys of same cardinality.
- (c) A vector clock in NoSQL environment showing conflict of concurrent execution.
- (d) Hybrid fragmentation of data in distributed databases.
- (e) A property graph in Neo4j showing a relationship.

2+2+2+2+2
